

Constipation in Parkinson's Disease: A Systematic Review and Meta-Analysis

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Keywords

Parkinson's disease · Prediction · Constipation · Risk · Association

Abstract

Introduction: Constipation is a common nonmotor symptom of Parkinson's disease (PD) and has been reported to increase the risk of developing PD. However, previous studies have yielded conflicting results. Understanding this correlation may promote early diagnosis and treatment of PD, which could help patients improve their quality of life. This study aimed to investigate the association between constipation and PD onset. **Methods:** The study was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and Meta-analyses of Observational Studies in Epidemiology (MOOSE) guidelines. We searched the Medline, Embase, Scopus, S1-NOMED, and Cochrane databases as well as specific journals from inception to September 2021 for observational studies that evaluated the association between constipation and the risk of PD. The Newcastle-Ottawa Scale was used to evaluate the methodological quality of the included studies. Associations were summarized as odds ratios (ORs) using a random-effects model. Subgroup, meta-regression, and sensitivity analyses were performed. **Results:** Seventeen studies comprising 3,024,193 participants (case-control = 1,636,831; cohort = 1,387,362) were eligible for inclusion. The pooled

OR for the association between constipation and PD was 2.36 (95% confidence interval: 1.93–2.88), although strong heterogeneity was observed ($I^2 = 90\%$, $p < 0.01$). Subgroup and meta-regression analyses indicated that study design and disease duration were the major sources of heterogeneity. A sensitivity analysis confirmed the stability of the outcomes. In addition, the prevalence of among those with prodromal PD was 20%, whereas it was only 11% in the control group ($p < 0.01$). Moreover, there were no significant age-based differences in constipation between the prodromal stage of PD patients and the controls ($p > 0.05$). **Conclusion:** Constipation has a relatively high incidence in the prodromal phase of PD and is associated with an increased risk of developing PD.

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Introduction

Parkinson's disease (PD) is a progressive neurodegenerative movement disorder with an incidence of approximately 1% among those over 60 years of age [1]. Nonmotor symptoms [2] such as anxiety, depression, sleep disorder, pain, and constipation are common in patients

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with PD. These nonmotor symptoms are often observed many years prior to the emergence of motor symptoms and exert a negative impact on patient quality of life [3, 4]. A better understanding of these early symptoms may aid in the identification of those at greater risk for PD, which may in turn improve the likelihood of treatment at an early stage.

The prevalence of constipation in patients with established PD ranges from 50 to 80% [5], affecting multiple aspects of daily living. Several studies have reported a strong correlation between constipation and PD risk. A 2016 meta-analysis including nine studies [6] reported an odds ratio (OR) of 2.27 (95% CI: 2.09–2.46) for the correlation between constipation and PD. However, this result has been controversial given the small sample size. In recent years, several larger cohort and case-control studies have been published, including a total of 3,024,193 cases. Including these cases may increase the credibility of the previous conclusions. Therefore, we aimed to investigate the association between constipation and PD onset via a meta-analysis of studies published through September 2021.

Materials and Methods

This meta-analysis of observational studies was performed to assess the association between constipation and subsequent diagnosis of PD. The study was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [7], the Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines [8], and the Cochrane Handbook for Systematic Reviews and Interventions. This study was registered in PROSPERO (CRD42021281795).

Search Strategy

We searched the PubMed (Medline), Embase, Scopus, SINOMED, and Cochrane libraries from database inception to September 24, 2021, using search terms related to “Parkinson’s Disease” and “Constipation.” The detailed search strategy is included in the Supplementary Data. We also manually searched relevant journals (e.g., *Neurology*, *Movement Disorders*, *NPJ Parkinson’s Disease*, *Journal of Parkinson’s Disease*, *Parkinsonism & Related Disorders*, *Parkinson’s Disease*), ChiCTR, and ClinicalTrials.gov using the aforementioned terms.

Eligibility Criteria

Studies were included if they (1) clearly diagnosed PD as an outcome; (2) clearly defined constipation based on reliable records such as prescriptions, structured interviews, or questionnaires; (3) presented data for ORs, relative risk (RR), hazard ratios (HRs), and their corresponding 95% CIs or provided enough data to compute these statistics; and (4) the study design was defined as observational (case-control and cohort). Studies with a high risk of bias (Newcastle-Ottawa Scale [NOS] scores ≤ 4) were excluded.

Data Extraction

Two reviewers independently extracted data from the full-text and supplementary materials using a predesigned data extraction form (online suppl. Table 1; for all online suppl. material, see www.karger.com/doi/10.1159/000527513). Disagreements were resolved through discussion. The extracted information included the name of the first author, publication year, study design, location of the study, sample size, disease duration, approaches to identifying cases of constipation and PD, adjusted risk estimates with corresponding 95% CIs, and other relevant study characteristics. The International Classification of Diseases codes, questionnaire results, and medical records were reviewed and abstracted by two reviewers to ascertain cases of constipation. For studies in which constipation was defined based on the frequency of bowel movements and laxative use, we selected the most conservative cutoff (bowel movement frequency of every other day or less or mild laxative use).

For studies that did not report effect sizes or confidence intervals, we calculated ORs using the total number of participants and constipation events in each group [9]. The OR value refers to the ratio of the exposed divided by the ratio of the unexposed. For the prevalence analysis, we extracted constipation cases in the prodromal PD group from relevant studies. In our analysis, prodromal PD was defined as the stage in which nonmotor symptoms were present without sufficient criteria for clinical diagnosis [10].

Quality Assessment

Study bias was assessed using the NOS, in accordance with the recommendations of the Cochrane Collaboration Group (2021). The NOS examines potential bias related to selection, comparability, and outcomes/exposures, with overall scores ranging from 0 to 9. Scores ≤ 4 indicate a high risk of bias, scores 5–6 indicate a moderate risk of bias, and scores ≥ 7 indicate a low risk of bias [11] (online suppl. Table 2).

Statistical Analysis

Given the low incidence of PD, all HRs and RRs were directly treated as ORs in our pooled analysis because the OR may approximate the RR and HR under the rare disease assumption [12, 13]. $I^2 > 50\%$ was used as the threshold for significant heterogeneity [14], and subsequent analyses were conducted using a random-effects model. Meta-regression models combined with subgroup analyses were used to investigate whether specific variables explained any of the heterogeneity among studies (<https://www.sciencedirect.com/topics/medicine-and-dentistry/meta-regression>). Subgroup analyses were conducted based on study type, location, source of the study population, disease duration (prodromal stage), literature quality, or method of diagnosis. We also performed sensitivity analyses by removing one study at a time and recalculating the risk to evaluate whether the overall results were markedly affected by those of a single study. We visually assessed publication bias using funnel plots and quantitatively using the AS-Thompson test. All results were considered significant at $p < 0.05$. All analyses were performed using Review Manager (version 5.4) and R Studio (version 4.03). The meta package (version 4.18–2) was used to produce the pooled estimates and forest plots. The metafor package (version 3.0–2) was used to conduct the meta-regression analysis.

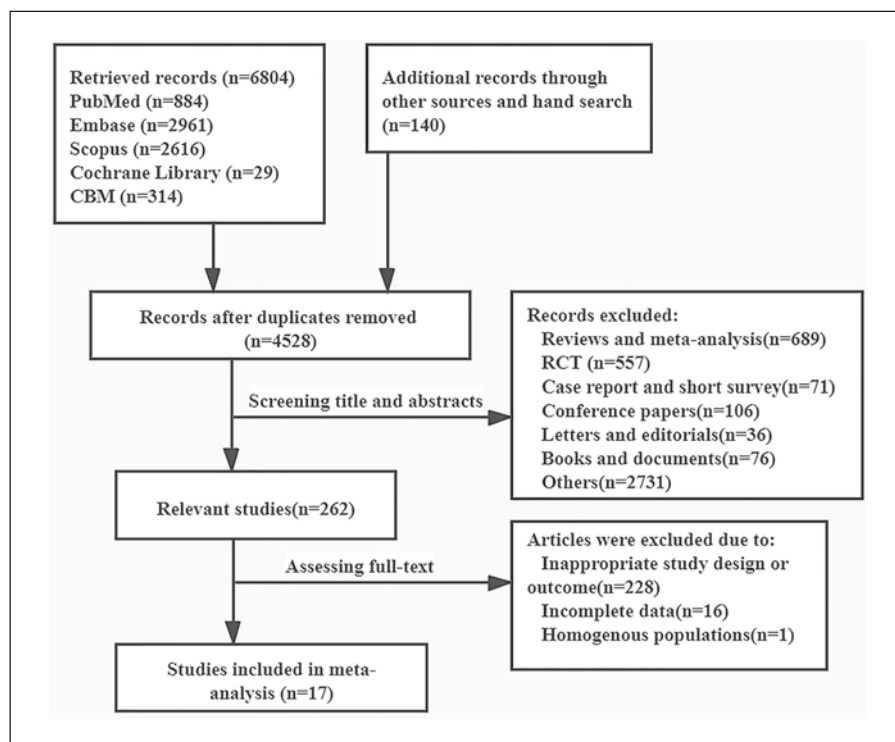


Fig. 1. PRISMA flow diagram for the literature search.

Results

The search strategy retrieved 6,944 records from the designated databases. After removing duplicate records, 262 potentially eligible articles were retained by screening titles and abstracts. After a full-text review, 17 studies [15–31] (11 case-control and 6 cohort) were included in the meta-analysis (Fig. 1).

Study Characteristics

The baseline characteristics and outcomes of the included studies are summarized in Table 1. There were 11 case-control studies and six cohort studies, with an aggregated total of 3,024,193 participants (case-control = 1,636,831; cohort = 1,387,362). The studies were published between 1997 and 2021. Seven studies were conducted in European countries, five were conducted in the USA, three were conducted in China, one was conducted in Mexico, and one was conducted in Argentina. The methods used to identify constipation varied across studies because the diagnosis generally originated from direct evaluation of established registers, medical records, and self-report questionnaires (frequency of bowel movements). A total of 15 studies were ranked as high quality

(score ≥ 7), while others were considered to be of moderate quality (score of 5–6), as shown in Table 1.

Constipation and Risk of PD

Of the 17 included studies, 13 reported a positive correlation between constipation and the onset of PD, another three studies did not report statistically significant correlations, and one reported a negative correlation. Figure 2 shows the results of the pooled analysis from the random-effects model. Despite the high heterogeneity of the included studies ($I^2 = 90\%$, $p < 0.01$), constipation was associated with an increased risk of PD (OR, 2.36; 95% CI, 1.93–2.88). Incidence rates for constipation in the prodromal phase of PD were calculated using data derived from three cohort studies (16, 30, and 29). The pooled prevalence of constipation in the prodromal PD group was 20% (95% CI: 0.09–0.34), while that in the control group was 11% (95% CI: 0.04–0.20) ($\chi^2 = 93.67$, $p < 0.01$), and the sensitivity analysis indicated the stability of prevalence estimates (online suppl. Fig. 1). There were no significant differences in age between the PD and control groups (60.9 years vs. 61.1 years; Student's t test, $p = 0.58$).

Table 2 shows that results did not differ substantially according to location, source of the study population, lit-

Table 1. Characteristics of included studies

Study	Country	Study type	Participants; age	Method of diagnosis		Results	Quality score
				PD	constipation		
Gonera and van't Hof, 1997 [15]	The Netherlands	Case-control	60 PD cases and 58 age- and sex-matched controls (hospital setting); age range 28–73 years	Review of hospital records using ICPC	ICPC	4 cases were seen in prodromal PD versus 8 cases in controls	7
Abbott et al., 2001 [16]	USA	Cohort	6,790 cases of men (general population); age range 51–75 years	Review of hospital records by published criteria	Frequency of bowel movement <1/day	Adjusted RR, 2.7; 95% CI, 1.3–5.5	8
Savica et al., 2009 [17]	USA	Case-control	196 PD cases and 196 controls matched by age and sex (general population); age range 41–97 years	Review medical records-linkage system of the Rochester Epidemiology Project	Clear diagnosis or concomitant use of laxative drugs in the medical records	Adjusted OR, 2.18; 95% CI, 1.32–3.61	8
Gao and Chen, 2011 [18]	USA	Cohort	33,901 men and 93,767 women (health professionals); NA	Review medical records	Self-report frequency of bowel movement, every 2 days or less	Adjusted RR, 3.93; 95% CI, 2.26–6.84	7
Cersosimo et al., 2013 [19]	Argentina	Case-control	129 PD cases and 120 controls were spouses of cases (hospital setting); age range 30–82 years	Medical records linkage diagnosis by UK Brain Bank criteria	Frequency of bowel movements <3/week	27 cases were seen in prodromal PD versus 10 cases in controls	5
Plouvier et al., 2014 [21]	The Netherlands	Case-control	86 PD cases and 78 controls matched by age, sex, and socioeconomic and marital status (hospital setting); median age in PD group 72.3 years	Cases from the CMR database	Medical records	Adjusted OR, 3.32; 95% CI, 1.1–10.4	6
Pont-Sunyer et al., 2014 [22]	Spain; Austria	Case-control	109 PD cases and 107 controls matched by age and gender (hospital setting); median age in PD group 69 years	Cases form the medical centers diagnosed by UK Brain Bank criteria	Questionnaire and self-report	Adjusted OR, 2.7; 95% CI, 1.4–5.2	8
Lin and Liu, 2014 [20]	China (Taiwan)	Cohort	551,234 cases (general population); median age was 43.5 years	Medical records linkage diagnosis using ICD-9-CM	The number of prescriptions for laxatives within 1 year	Adjusted RR, 2.42; 95% CI, 2.06–2.85	7
Schrag and Horsfall, 2015 [23]	UK	Case-control	8166 PD cases and 46,755 controls matched by age, sex, and registration period (hospital setting); age range 68–81 years	Medical records using read codes	Prescriptions for constipation	Adjusted OR, 2.01; 95% CI, 1.62–2.49	8
Wu and Liao, 2015 [24]	China (Taiwan)	Case-control	705 PD cases and 2,820 age-, sex-, and index-date- matched controls (hospital setting); median age 66 years	Medical records linkage diagnosis using ICD-9-CM	Based on ICD-9-CM and prescriptions for constipation	285 cases were seen in prodromal PD versus 678 cases in controls	7
Svensson and Henderson, 2016 [25]	Danish	Cohort	31,905 patients with constipation and 159,092 controls matched by year of birth, sex, and year of index date (general population); median age 62 years	Medical records linkage diagnosis using ICD-10	Prescriptions for constipation	Adjusted RR, 3.03; 95% CI, 2.50–3.66	8

Table 1 (continued)

Study	Country	Study type	Participants; age	Method of diagnosis		Results	Quality score
				PD	constipation		
Rodríguez-Violante and Saráchaga, 2017 [26]	Mexico	Case-control	430 PD cases and 430 controls matched by gender and age (hospital setting); median age 66.3 years	Medical records linkage diagnosis using MDS-UPDRS	Medical records	Adjusted OR, 1.39; 95% CI, 0.98–1.96	8
Hughes et al., 2018 [27]	USA	Case-control	120 PD cases and 6,479 controls matched by gender and age (health professionals); median age 73 years	Self-reported diagnosis of PD	Bowel movement frequency of every other day or less and/or laxative use at least twice a week	Adjusted OR, 4.93; 95% CI, 3.36–7.24	8
Jacobs et al., 2020 [28]	UK	Case-control	1,276 PD cases and 500,406 unmatched controls (general population); age range 40–69 years	Self-reported or medical records linkage diagnosis using ICD	Health records	Adjusted OR, 1.28; 95% CI, 0.32–5.13	7
Yang et al., 2020 [30]	China	Cohort	510,134 cases (general population); median age 52 years	Medical records linkage diagnosis using ICD-10	Bowel movement frequency of every other day or less	Adjusted RR, 2.13; 95% CI, 1.74–2.60	7
Marini et al., 2020 [29]	Germany	Cohort	86 cases and 453 matched controls (general population); median age 69.1 years	Medical history linkage diagnosis using UPDRS-III	Medical records	5 cases were seen in prodromal PD versus 15 cases in controls	8
Scott and Lim, 2021 [31]	USA	Case-control	303,693 PD cases and 1,214,772 controls matched by gender, age, smoking status, race, ethnicity, and birth year (general population); median age 74.6 years	Medical records linkage diagnosis using ICD	ICD-9 and ICD-10 diagnosis	Adjusted OR, 1.57; 95% CI, 1.55–1.60	8

ICPC, the International Classification of Primary Care; NA, not reported; ICD, International Statistical Classification of Diseases and Related Health Problems; UPDRS, Unified Parkinson's Disease Rating Scale; OR, odds ratio; RR, relative risk; HR, hazard ratio; PD, Parkinson's disease; CMR, Continuous Morbidity Registration.

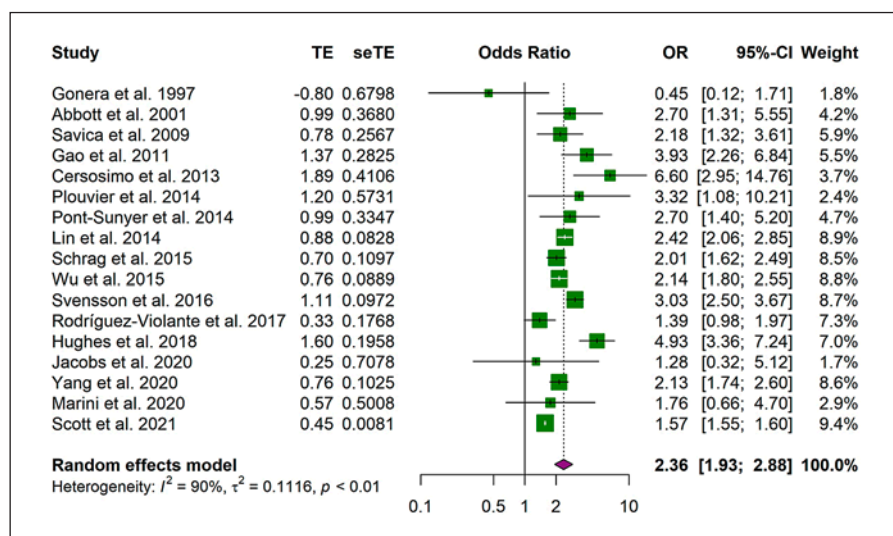


Fig. 2. Forest plot for the association between constipation and PD onset.

Table 2. Stratified subgroup analysis

	No	HR [95% CI]	Heterogeneity		Meta-regression
			I^2 (%)	p value	p value
Study type					
Case-control	11	2.21 [1.72–2.83]	86	0.01	0.01
Cohort	6	2.57 [2.17–3.05]	48	0.09	
Location					
USA	5	2.78 [1.57–4.92]	92	0.01	0.51
Europe	6	2.10 [1.44–3.06]	68	0.01	
China	3	2.24 [2.03–2.49]	0.0	0.50	
Others	3	2.75 [1.14–6.63]	85	0.01	
Source					
Hospital setting	7	2.11 [1.57–2.83]	69	0.01	0.11
General population	8	2.18 [1.65–2.87]	92	0.01	
Others	2	4.58 [3.34–6.28]	0.0	0.51	
Disease duration, years					
≥10	11	2.06 [1.61–2.63]	87	0.01	0.01
5–10	3	2.43 [1.97–3.00]	54	0.11	
≤5	3	5.00 [3.59–6.97]	0.0	0.61	
Study quality score					
≥7	15	2.25 [1.84–2.75]	91	0.01	0.06
<7	2	5.23 [2.72–10.06]	0.0	0.33	
Methods of diagnosis					
Frequency of bowel movement	5	3.60 [2.24–5.78]	81	0.01	0.56
Medical record	7	2.20 [1.77–2.74]	68	0.01	
Others	5	1.80 [1.36–2.38]	78	0.01	
Use of laxatives					
Yes	6	2.61 [2.08–3.27]	76	0.01	0.26
Not	11	2.14 [1.62–2.84]	87	0.01	

No, number of studies.

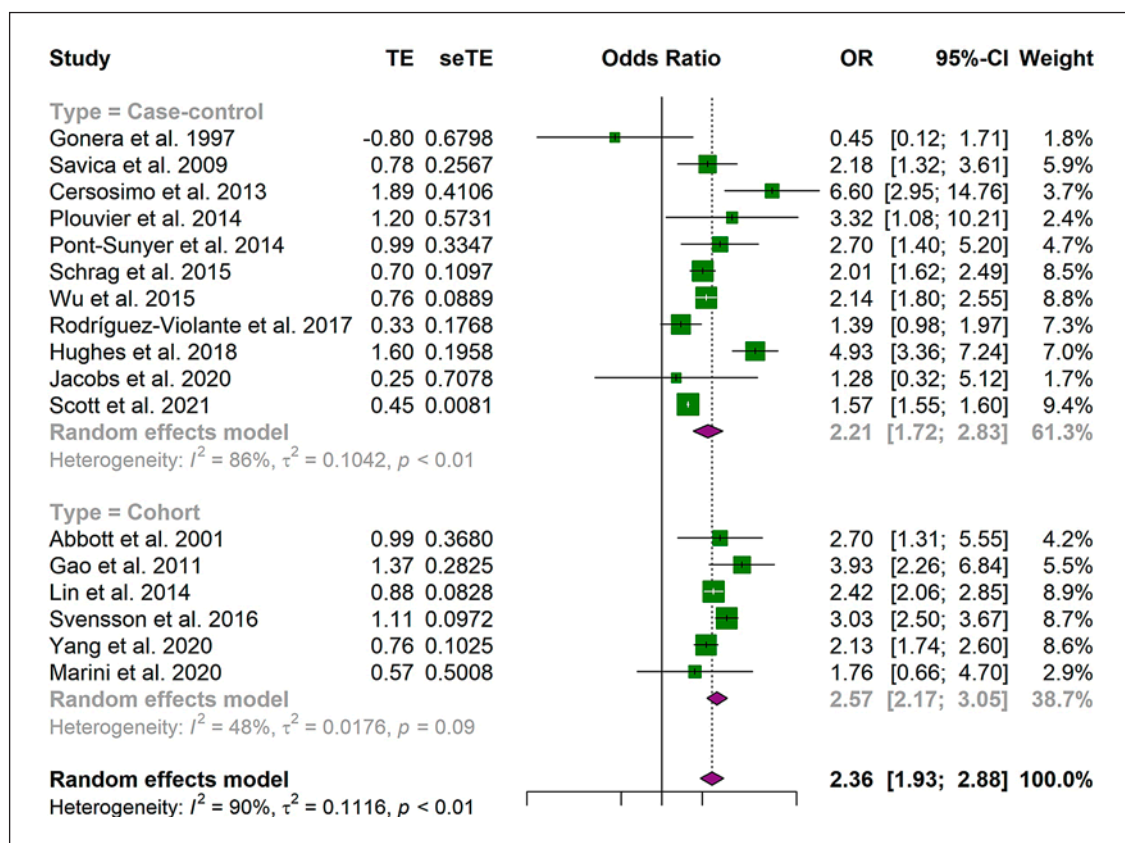


Fig. 3. Forest plot for the subgroup analysis according to study type.

erature quality, or diagnostic methods. Similar results were obtained in the meta-regression analysis. In both the case-control and cohort groups, we observed a positive association between constipation and PD, although the heterogeneity mainly existed in the case-control group (Fig. 3). Further subgroup analysis was performed based on disease duration in the case-control group, after excluding the study with the greatest weight [31]. In this analysis, we observed an increased risk of PD (more than 10 years: OR, 1.90; 95% CI: 1.55–2.32; $I^2 = 45\%$, $p = 0.09$; less than 5 years: OR: 5.00; 95% CI: 3.59–6.97; $I^2 = 0\%$, $p = 0.61$), and heterogeneity across studies was eliminated (Fig. 4). Taken together, these results suggest that differences in disease duration are another major source of heterogeneity in studies of constipation and PD risk.

Sensitivity Analysis

To investigate the influence of a single study on the overall risk estimate, sensitivity analyses were performed by excluding one study at a time. Figure 5 shows that the

combined ORs for overall risk were consistent and without obvious fluctuation, ranging from 2.23 (95% CI: 1.84–2.70) to 2.46 (95% CI: 2.07–2.92). This outcome demonstrates the robustness of our analyses.

Evaluation of Publication Bias

Visual inspection of the funnel plot revealed a near-symmetrical distribution (online suppl. Fig. 4). We calculated publication bias using the AS-Thompson test, which introduced between-study heterogeneity into the regression weight [32]. No obvious evidence of publication bias was detected (AS-Thompson test, $p = 0.8135$).

Discussion

In this study, we systematically analyzed the correlation between constipation and PD risk via a meta-analysis of 17 studies involving 3,024,193 cases. The pooled OR for the association between constipation and PD was 2.36 (95% CI,

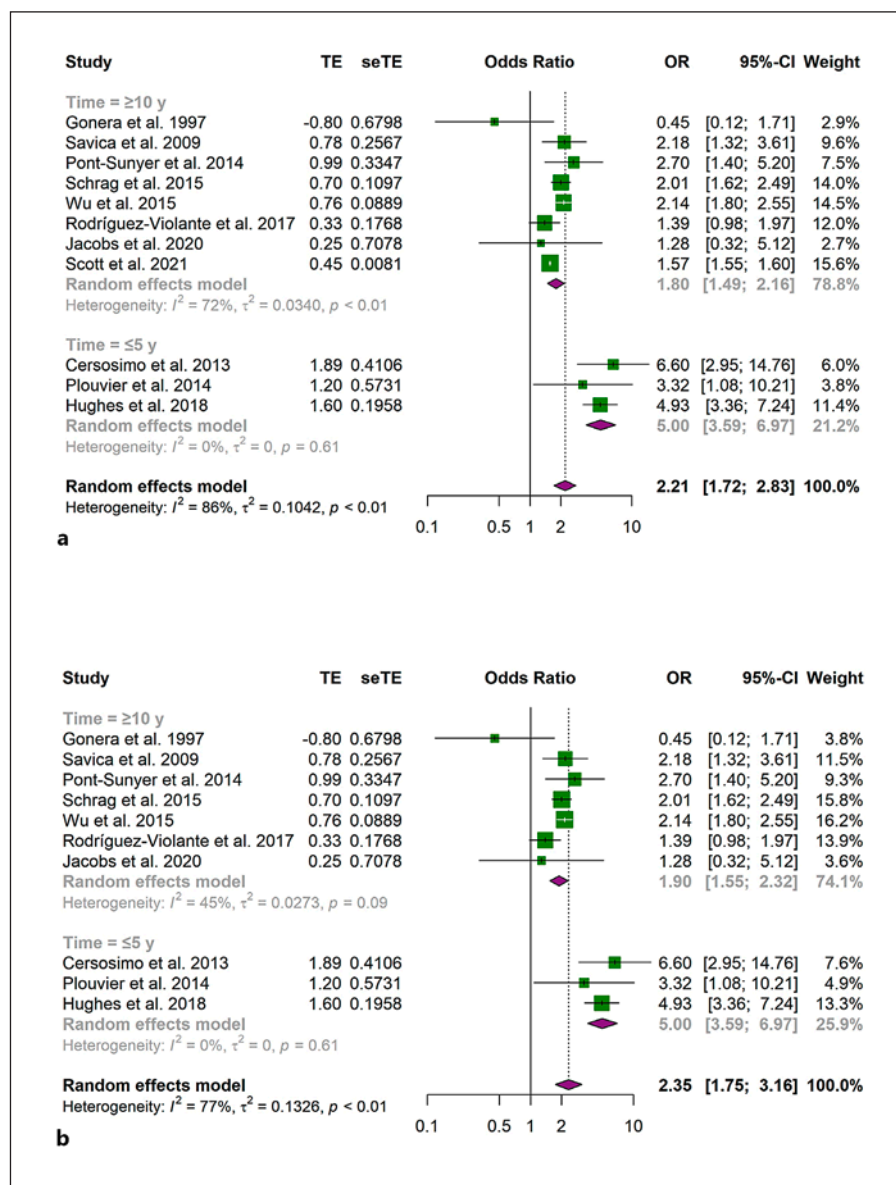


Fig. 4. Forest plot for the subgroup analysis in the case-control group. **a** Forest plot for subgroup analysis according to disease duration. **b** Forest plot from (a) after excluding the study with the greatest weight.

1.93–2.88, $I^2 = 90\%$). Our results are similar to those of a previous study [6]. Considering the different methods used to extract the data, we re-extracted the adjusted effect sizes for overlapping studies for a pooled analysis, demonstrating the reliability of our findings. However, significant heterogeneity was observed across studies. Subgroup analysis revealed that study type and duration were the main sources of heterogeneity, although the type of study did not contribute to significant heterogeneity in previous studies. The relatively small sample size (nine studies) and racial disparities in the population may explain the inconsistency of these conclusions. Our study included multiple popula-

tions and a larger sample size, thus providing a better representation of the general population.

Another subgroup analysis indicated that heterogeneity mainly existed in the case-control group ($I^2 = 86\%$), with only slight heterogeneity in the cohort group ($I^2 = 48\%$), suggesting that research methods contributed to heterogeneity. The case-control design is a retrospective method that can be associated with memory bias as the longer time span implies that more information may be lost. Therefore, we performed a subgroup analysis based on the duration of the prodromal phase. In studies with a duration of <5 years, I^2 was 0%. In contrast, I^2 was 45%

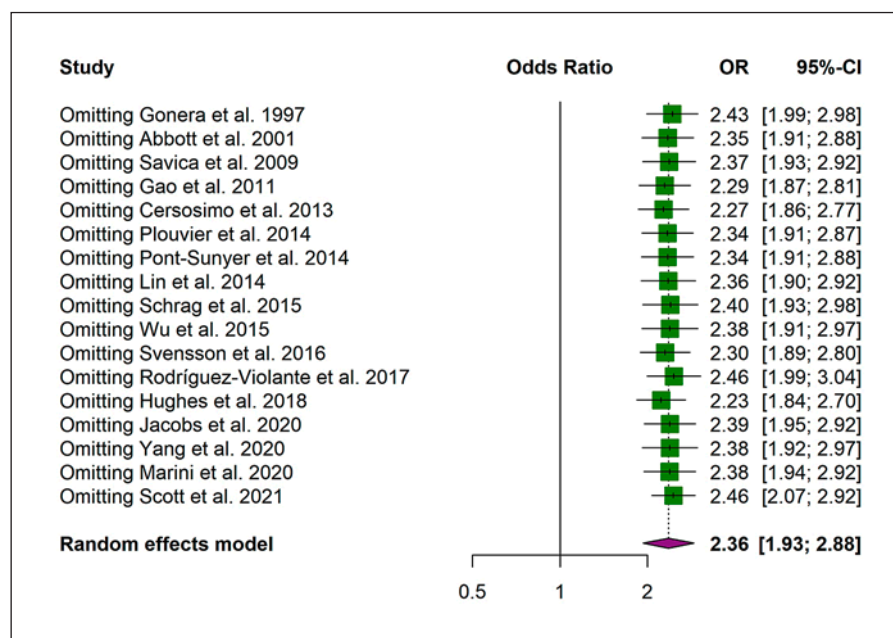


Fig. 5. Sensitivity analysis.

for studies with a duration of >10 years. This significant increase in heterogeneity highlights disease duration as an important confounding factor that can lead to memory bias. Moreover, the sample sizes of case-control studies are generally small, meaning that the sample may not be representative of the general population. Selection bias in such studies may also increase heterogeneity. A cohort study is often considered more reliable given the reduced impact of memory and selection bias, which may decrease overall heterogeneity in the pooled analysis.

Subsequently, we conducted a sensitivity analysis to further analyze the robustness and reliability of our conclusions. Indeed, the results were relatively stable, with overall ORs ranging from 2.23 (95% CI: 1.84–2.70) to 2.46 (95% CI: 2.07–2.92). This result verifies the relative robustness of our findings and emphasizes that heterogeneity was not introduced by one specific study.

In the analysis of publication bias, six studies did not fall within the field of the 95% CI in the funnel plot, suggesting that asymmetry may have been caused by study heterogeneity. This is also consistent with the abovementioned result, suggesting that heterogeneity was derived from the study type and duration. Due to the high degree of heterogeneity among the included studies, we considered it more reasonable to use the AT-Thompson test rather than the funnel plot to assess publication bias. This test verified that there was no significant publication bias.

We evaluated the quality of the included studies using the NOS, which is widely applied for meta-analysis. NOS

scores for the included studies ranged from 5 to 9. Poorly representative samples and unjustified methods of determining exposure were the key areas contributing to lower scores. On one hand, selection by doctors or nurses can lead to selection bias. However, some studies simply chose disease codes based on medical records or whether laxatives were used to define exposure to “constipation,” which is not a reliable strategy. Combining the disease code with medical history indicating drug treatment is a more accurate criterion for determining exposure to “constipation.”

Two studies reported an association between sex and the risk of developing PD. Our analysis indicated that male sex was more strongly associated with risk in patients with constipation, although this must be verified in future studies (online suppl. Fig. 2). Additionally, our results suggest that the frequency of bowel movements and the severity of constipation are positively correlated with the risk of developing PD. Given that only three included studies explored these issues, a precise conclusion could not be drawn (online suppl. Fig. 3).

Because constipation can occur decades before the emergence of motor symptoms of PD, analyzing this association may promote early diagnosis and treatment of PD. Constipation mainly manifests as reduced frequency of or difficulty in defecation, specifically due to weakening of colon peristalsis, slowing of propulsion speed, and prolonged presence of stool in the intestinal tract. Brak et al. [33] argued that PD may originate from the periph-

eral gastrointestinal tract [33, 34] as pathogenic microorganisms encroach on the intestine, leading to abnormal folding of α -synuclein [35]. These structurally abnormal α -synuclein aggregates exhibit prion-like propagation, invading peripheral tissues, and migrating along the vagus nerve to infect the substantia nigra via the nigro-vagal pathway. This cascade may produce a “snowball” effect via the combination of oxidative stress, inflammation, and other factors [36–38], eventually promoting the apoptosis of dopaminergic neurons. Several other studies have reported that constipation may be related to the formation and accumulation of α -synuclein in the gut [39–41], although the pathological mechanism underlying this phenomenon remains to be clarified. Given these findings, the associations between gastrointestinal dysfunction and PD and the search for gut biomarkers of PD have become hot topics in recent years. While α -synuclein is likely among the most promising of these potential markers, neuroimaging markers such as dopamine transporter scanning and metaiodobenzylguanidine myocardial scintigraphy may be useful for diagnosing prodromal PD in patients with bowel symptoms [42]. Large-scale studies are required to verify the clinical value of these markers. Constipation may be an early manifestation of PD. However, the present results suggested that constipation is more like the potential risk factor of PD. Thus, constipation may be one of the nonmotor symptoms in PD as well as a risk factor.

Our study had some limitations. First, we only searched for studies published in Chinese or English, which may have excluded some non-English studies. Second, we only chose general biomedical and science electronic databases for the literature search. Although we also searched professional magazines such as *Movement Disorders*, we cannot guarantee that all relevant studies were included. To minimize literature omissions, we designed the search strategy to be as specific as possible by developing a search “equation” with a detailed list of MeSH terms and text words, such as *Parkinson’s disease*, *parkinsonism*, or *Lewy body*. There were also inconsistencies in the management of laxatives among studies. Laxative use was defined as constipation or used to represent the severity of constipation in some studies, while other studies directly excluded cases involving laxative use. Although some studies also adjusted their analyses for laxative use, others did not. Furthermore, either the disease codes or clear descriptions of constipation were used as evidence of “exposure.” The variety of definitions for constipation in different countries and territories may have affected our results.

Conclusion

Constipation is a common nonmotor symptom that occurs in the early stages of PD and has been shown to increase the risk of PD onset. Our analysis indicated that patients with constipation had a 2.36-fold increased risk of PD when compared with those without constipation, which may last for decades. Considering the irreversible course of PD, early diagnosis and intervention are critical. Combining constipation with other nonmotor symptoms such as anosmia, rapid eye movement sleep behavior disorder, and other makers may aid in predicting the occurrence of PD.

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Statement of Ethics

This article was based on data from previously published studies, and therefore, ethical approval was not required.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

Lulu Yao and Xiaobo Huang: conception and design. Lulu Yao, Wei Liang, and Qian Wang: collection and assembly of data. Lulu Yao, Qian Wang, and Jiahao Chen: data analysis and interpretation. Lulu Yao and Wei Liang: drafting the article. Xiaobo Huang: critically revising the article. All authors contributed to the article and approved the submitted version.

Data Availability Statement

The original contributions presented in the study are included in the article and its online supplementary material. Further inquiries can be directed to the corresponding author.

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